Experiment 03 – Operators

Learning Objectives: To understand the method and usage of operators in a program.

Problem Statement :

1: Write a Java program to demonstrate the use of arithmetic, relational, logical, assignment, and unary operators. Explain the output of each operation with appropriate comments.

Theory:

Operators in Java and its Types:

1. Arithmetic Operators
2. Logical Operators
3. Relational Operators
4. Unary Operators
5. Assignment Operators

Arithmetic Operators in Java:

Arithmetic Operators are used to performing mathematical operations like addition, subtraction, etc. Assume that A = 10 and B = 20 for the below table.

Operator	Description	Example
+ Addition	Adds values on either side of the operator	A+B=30
– Subtraction	Subtracts the right-hand operator with left-hand operator	A-B=-10
* Multiplication	Multiplies values on either side of the operator	A*B=200
/ Division	Divides left hand operand with right hand operator	A/B=0
% Modulus	Divides left hand operand by right hand operand and returns remainder	A%B=0

Logical Operators in Java:

Logical operators in Java are used to combine or modify boolean expressions, resulting in a boolean value (true or false).

Operator	Description	Example
&& (and)	True if both the operands is true	a<10 && a<20
|| (or)	True if either of the operands is true	a<10 || a<20
! (not)	True if an operand is false (complements the operand)	!(x<10 && a<20)

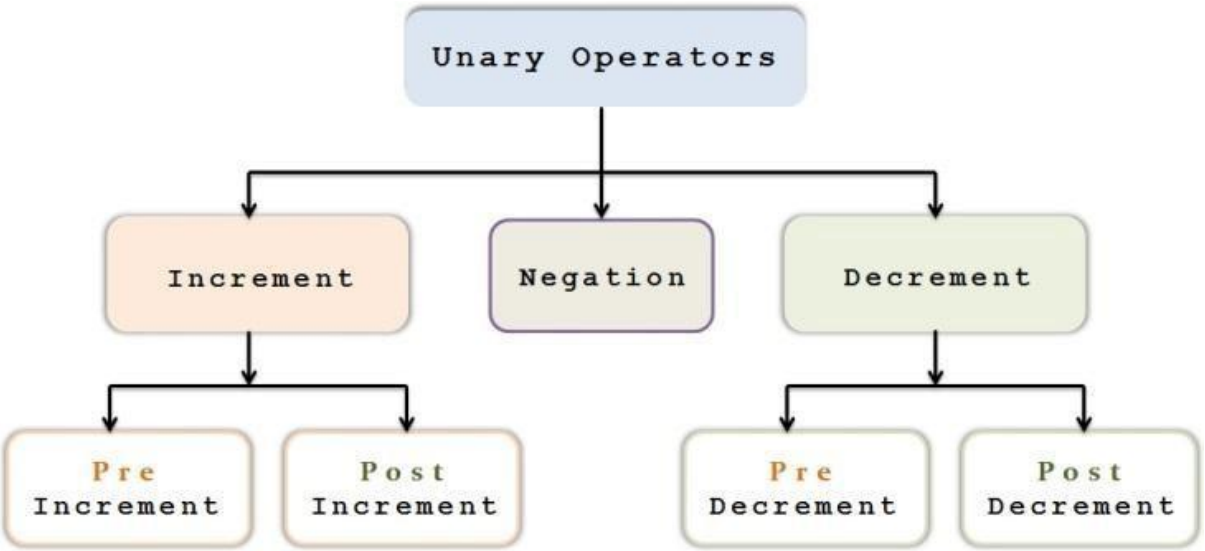
Relational Operators in Java:

Relational operators in Java are used to compare two operands and determine the relationship between them. They always return a boolean value, either true or false.

Operation	Operator	Description
Equal To	==	Compares the values of two variables for equality
Not Equal To	!=	Compares the values of two variables for inequality
Greater Than	>	Checks if one value is greater than the other value
Lesser Than	<	Checks if one value is lesser than the other value
Greater Than Or Equal To	>=	Checks if one value is greater than or equal to the other value
Lesser Than Or Equal To	<=	Checks if one value is lesser than or equal to the other value

Unary Operators in Java:

The unary operators works on only one operand, they perform various operations such as incrementing/decrementing a value by one, negating an expression, or inverting the value of a boolean.



Assignment Operators in Java:

Assignment operators in Java are used to assign a value to a variable. They modify the value of a variable based on the operation performed.

Operator	Description	Example
=	assigns values from right side operand to left side operand	x = y
+ =	adds right operand and left operand then assign the result to left operand	x + = y is same as x = x + y
- =	subtracts right operand from the left operand and assign the result to left operand	x - = y is same as x = x - y
* =	multiply left operand with the right operand and assign the result to left operand	x * = y is same as x = x * y
/ =	divides left operand with the right operand and assign the result to left operand	x / = y is same as x = x / y
% =	calculate modulus by dividing left operand by right operand and assign the reminder to left operand	x % = y is same as x = x % y

Problem Statement 1: Write a Java program to demonstrate the use of arithmetic, relational, logical, assignment, and unary operators. Explain the output of each operation with appropriate comments.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

D:\ritik_cse_cs_A>java Operator.java
=== Arithmetic Operators ===
a + b = 16
a - b = 8
a * b = 48
a / b = 3
a % b = 0

=== Relational Operators ===
a == b : false
a != b : true
a > b : true
a < b : false
a >= b : true
a <= b : false

=== Logical Operators ===
x && y : false
x || y : true
!x : false

=== Assignment Operators ===
c += 5 : 15
c -= 3 : 12
c *= 2 : 24
c /= 4 : 6
c %= 5 : 1

=== Unary Operators ===
Initial value of d: 5
++d : 6
d++ : 6
After d++ : 7
--d : 6
-d : -6

D:\ritik_cse_cs_A>
```

Learning Outcomes: Students will able to

LO1.1 Understand the primitive and non-primitive data types.
LO1.2 Understands the different operators.
LO1.3 Understands about Reference Object.

Course Outcome:

CO1: Understand about Various operators.

Conclusion:

For Faculty Use only:

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	Total
Marks Obtained				